

Project Title: Cloud-Based Video Streaming Platform using Flask and Google Cloud Storage

Abstract:

This project demonstrates a simple yet functional video streaming platform where users can upload and view videos directly from the browser. The backend is built using Python with Flask, while Google Cloud Storage (GCS) handles the media storage. The web interface is styled using CSS for a cleaner UI experience. This platform showcases how cloud services can be integrated to handle scalable storage and dynamic content delivery, making it ideal for educational or prototype use cases. The system allows ano...

Features:

- Upload video files to a GCS bucket
- Stream video content securely using signed URLs
- Simple user interface without login requirements
- Lightweight Flask backend

```

# app.py (Backend)
from flask import Flask, render_template, request, redirect, url_for, flash
from google.cloud import storage
import os

os.environ['GOOGLE_APPLICATION_CREDENTIALS']
'video-streaming-project-454118-c030128fea48.json'

app = Flask(__name__)
app.secret_key = 'your_secret_key'
storage_client = storage.Client()
bucket = storage_client.get_bucket('video-streaming-bucket-123')

@app.route('/')
def index():
    blobs = bucket.list_blobs()
    videos = [blob.name for blob in blobs if blob.name.endswith('.mp4')]
    return render_template('index.html', videos=videos)

@app.route('/upload', methods=['POST'])
def upload():
    video = request.files['video']
    if video:
        blob = bucket.blob(video.filename)
        blob.upload_from_file(video)
        flash('Video uploaded successfully.')
    return redirect(url_for('index'))

@app.route('/play/<video_name>')
def play(video_name):
    blob = bucket.blob(video_name)
    if not blob.exists():
        flash('Video not found')
        return redirect(url_for('index'))
    video_url = blob.generate_signed_url(expiration=3600)
    return render_template('play.html', video_url=video_url, filename=video_name)

```

```
<!-- templates/index.html -->
<!DOCTYPE html>
<html>
<head>
    <title>Video Streaming Platform</title>
    <link rel="stylesheet" href="/static/style.css">
</head>
<body>
    <div class="container">
        <h1>Video Upload</h1>
        <form action="/upload" method="post" enctype="multipart/form-data">
            <input type="file" name="video" required>
            <button type="submit">Upload</button>
        </form>
        <h2>Available Videos</h2>
        <ul>
            {% for video in videos %}
                <li><a href="/play/{{ video }}">{{ video }}</a></li>
            {% endfor %}
        </ul>
    </div>
</body>
</html>
```

```
<!-- templates/play.html -->
<!DOCTYPE html>
<html>
<head>
    <title>Playing Video</title>
</head>
<body>
    <h1>Now Playing: {{ filename }}</h1>
    <video width="720" controls>
        <source src="{{ video_url }}" type="video/mp4">
        Your browser does not support video.
    </video>
    <br><a href="/">Back to Home</a>
</body>
</html>
```

```
/* static/style.css */
body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 20px;
    background-color: #f4f4f4;
}
.container {
    background: white;
    padding: 20px;
    border-radius: 8px;
}
h1, h2 {
    color: #333;
}
ul {
    list-style-type: none;
    padding: 0;
}
li {
    margin: 10px 0;
}
```